

Logistic Regression Equivalent Weights for Survey Inference: Construction and Asymptotic Properties

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Abstract

Equivalent weights from regression models provide a bridge between design-based and model-based survey inference. We develop a frequentist survey-weighting framework for logistic regression equivalent weights under categorical poststratification. The resulting weights are defined as the scaled first derivatives of the model-based population estimator with respect to the observed outcomes and admit an explicit closed-form expression in terms of the fitted logistic model and population cell structure. We establish convergence of the effective sample size ratio, asymptotic linearity of the resulting weighted estimator, and a consistent plug-in variance estimator under a finite-cell superpopulation framework. Simulation results examine population recovery, regression performance, ESS, and the finite-sample behavior of the asymptotic variance approximation. An application to the Fragile Families and Child Wellbeing Study illustrates how the proposed weighting construction can be used in an empirical survey setting. We emphasize the relationship between logistic equivalent weighting and recent work on locally equivalent weights for nonlinear multilevel regression and poststratification, while focusing specifically on survey-weight construction, ESS behavior, and frequentist inference.

1 Introduction

Survey weights are central to making sample data representative of target populations. Constructing base weights or adjusting them involves multiple steps and choices, often relying on auxiliary information to align estimates with census counts or other population benchmarks. Survey data also frequently face missingness and nonresponse, which can complicate these procedures and contribute to unstable or highly variable weights.

A substantial body of literature addresses nonresponse adjustment, design- and model-based inference, and small area estimation (Chapman et al., 1986; Little, 1986; Bethlehem et al., 1996; Chu and Goldman, 1997; Lu and Gelman, 2003; Little, 2015; Rao and Molina, 2015; Haziza and Beaumont, 2017; Skinner and Wakefield, 2017; Chen et al., 2017; Liu et al., 2023). At the same time, concerns remain regarding the construction of base weights, particularly when detailed population information is unavailable or confidential. The role of weights in regression is also an active methodological question: weighting does not automatically reduce bias, and calibration procedures such as raking can produce highly variable weights (Little and Rubin, 1987; Deville and Särndal, 1992; Miller, 2011).

Traditional approaches to constructing base weights follow design-based principles (Chu and Goldman, 1997; Valliant et al., 2013; Valliant and Dever, 2018). Model-based alternatives include regression-based equivalent-weight constructions, most notably the framework discussed by Gelman (2007). For linear models, the poststratified model-based estimator can be written exactly

as a weighted average of the observed outcomes. For nonlinear models, such an exact representation is generally unavailable, but a local linearization can produce an approximate weighting representation.

Recent work by Giordano et al. (2026) develops locally equivalent weights for multilevel regression and poststratification (MrP), including nonlinear outcome models such as logistic regression. Their framework derives an asymptotic closed-form expression for logistic locally equivalent weights from a Bayesian MrP formulation using Bernstein–von Mises and delta-method approximations. Our derivation proceeds independently from a frequentist perspective, starting from the logistic estimating equations and applying the chain rule to the model-based population estimator. Under categorical poststratification, this leads to the same closed-form weighting expression. Our focus, however, is on the interpretation and use of these weights as survey weights and on their frequentist sampling properties. We therefore study their effective sample size, asymptotic linearity, plug-in variance estimation, and finite-sample behavior.

The goal of this paper is to develop and evaluate logistic regression equivalent weights for survey inference. We compare design-based, linear, and logistic weighting through simulation and an empirical application, while establishing asymptotic results for the proposed logistic weights. The main theoretical results show that the ESS ratio converges to a population limit and that the weighted estimator admits an asymptotic linear representation with a consistent plug-in variance estimator under the stated regularity conditions. The empirical analysis then examines how these properties translate into finite-sample behavior.

The paper is organized as follows. Section 2 describes the target population, design-based weighting, and the model-based weighting constructions. Section 3 gives the asymptotic framework and theoretical results. Section 4 presents simulation and FFCWS results. Section 5 discusses the implications and limitations of the proposed approach. Detailed calculations for the local linearization of the logistic estimator are provided in Appendix A, while proofs of the asymptotic results are given in Appendix B.

1.1 Target Study Population and Design

The target survey population consists of live births occurring in large U.S. cities with populations over 200,000 between 1998 and 2000. This focus is motivated by the process of constructing base

weights for the Future of Families and Child Wellbeing Study (FFCWS). The FFCWS sample follows a stratified multistage design with 4,898 children, oversampling births to unmarried mothers at a ratio of 3 to 1, with the inclusion of a large number of Black, Hispanic, and low-income families (Reichman et al, 2001). Follow-up interviews were conducted across seven waves, when children were approximately ages 1, 3, 5, 9, 15, and 22. In constructing the FFCWS weights, four demographic variables were used for poststratification, and geographic information was incorporated to estimate population birth counts using the Centers for Disease Control and Prevention (CDC) annual natality data. These variables are mother’s marital status, race/ethnicity, age, education, and city of birth (Carlson,2008). To mirror the FFCWS study, we generate data and construct base weights in our simulation analysis that replicate the characteristics of the FFCWS design.

In large national surveys, stratified multistage cluster sampling (SMCS) is a common approach because it balances logistical challenges with the need to obtain sufficiently precise estimates for key subgroups. In this context, clusters, also referred to as primary sampling units (PSUs), are often used to simplify fieldwork. For example, the stratum is the city and the PSU is the hospital for cities selected with certainty in the FFCWS study. A simpler alternative is stratified simple random sampling (SRS), in which each unit in the population is assigned a group label and SRS is conducted within each group. These groups typically correspond to demographic strata or geographic regions.

In our simulation study, we choose to construct design-based weights using stratified simple random sampling (SRS) rather than stratified multistage cluster sampling (SMCS). While SMCS is widely used, intra-cluster correlation reduces efficiency by limiting the independent information gained from additional sampled units within the same cluster. This leads to higher standard errors and lower overall precision. Moreover, we expect minimal differences in weight estimation between SMCS and SRS once raking is applied. For our experiments, we draw samples of size $n = 3000$ from a finite population of size $N = 1,000,000$, chosen to mirror realistic survey conditions where the sample represents only a small fraction of the population.

1.2 Design-based Weighting

For a finite population and a probability sample, each unit i has a known selection probability $\pi_i > 0$, and the *base weight* is defined as

$$w_i = \frac{1}{\pi_i}.$$

A probability sample is realized under the following conditions: the set of all possible samples that can be selected from the finite population is well-defined under the sampling design, and each possible sample is associated with a known probability of selection. Moreover, every unit in the target population has a nonzero probability of being selected, with selection occurring via a random mechanism (Valliant and Dever, 2018). In our simulation study, the selection probability π_i is determined based on factors such as city of birth and demographic characteristics. These factors are used to form cells, where units within the same cell share identical probabilities of selection.

We then apply two standard adjustments: (i) nonresponse adjustment, where weights are multiplied by the inverse of the weighted response rate within adjustment cells (e.g., within each city) to account for survey nonrespondents; and (ii) raking, where weights are adjusted so that weighted sample counts align with known population margins. Nonresponse adjustments renormalize the weights to sum to the total population size, while raking aligns marginal totals without requiring full cross-classification, thereby avoiding the problem of empty cells. Given the extensive literature on nonresponse adjustments, we do not explore them in detail here. For recent applications to the FFCWS data, including a state-of-the-art two-stage approach using an optimally balanced Gaussian process, see (Vegetabile et al, 2020), refer to Lee and Gelman (2024).

Table 1 summarizes base weights from a representative simulation.

Table 1: Summary of base weights (simulation example)

	Min	1st Quantile	Median	Mean	3rd Quantile	Max
Value	2.86	121.40	260.80	333.33	492.49	1396.96

1.3 Estimating Population Mean from a Sample

Let X denote the poststratification variables whose joint distribution in the population is known, together with the survey response of interest y that we aim to estimate for the population. The possible strata of X form poststratification cells, denoted by s , with population sizes N_s and sample sizes n_s . The total population size is $N = \sum_{s=1}^S N_s$ and the total sample size is $n = \sum_{s=1}^S n_s$.

The population mean of y is

$$\theta = \frac{\sum_{s=1}^S N_s \theta_s}{N},$$

where θ_s is the population mean of y within cell s . The corresponding sample-based poststratified estimator is

$$\hat{\theta}^{PS} = \frac{\sum_{s=1}^S N_s \hat{\theta}_s}{N}.$$

For a probability sample, the Horvitz–Thompson (HT) estimator is

$$\bar{y}_{HT} = \frac{\sum_{i=1}^n w_i y_i}{N},$$

where $w_i = 1/\pi_i$ is the design weight. The Hájek estimator is

$$\bar{y}_H = \frac{\sum_{i=1}^n w_i y_i}{\hat{N}}, \quad \hat{N} = \sum_{i=1}^n w_i.$$

When the weights are normalized so that $\hat{N} = N$, the two expressions coincide algebraically. We use this normalization when comparing weighted estimators on the population-mean scale. Exact unbiasedness is a property of the Horvitz–Thompson estimator under the probability-sampling design; the Hájek estimator is generally a ratio estimator and need not be exactly unbiased in finite samples.

The poststratified estimator can be written as a weighted estimator when the cell-specific estimates $\hat{\theta}_s$ are represented through a regression or other model. We use this relationship to construct equivalent unit weights for the model-based methods below.

2 Model-based Weighting Methods

2.1 Linear Regression Weight

Let X denote the $n \times k$ design matrix of auxiliary variables for the samples, and let X^{pop} denote the $S \times k$ design matrix for the population poststratification cells. Here, n is the sample size, S is the number of poststratification cells, and k is the number of auxiliary variables (e.g., demographic or geographic characteristics) used for weighting. If the survey response of interest y has a linear relationship with the raking variables, then

$$\hat{\theta}_s = X_s^{pop} \hat{\beta}, \quad \hat{\beta} = (X^\top X)^{-1} X^\top y,$$

where $\hat{\beta}$ are estimated regression coefficients. The poststratified estimate of the population mean is

$$\hat{\theta}^{PS} = \frac{1}{N} \sum_{s=1}^S N_s (X_s^{pop} \hat{\beta}) = \frac{1}{N} (N^{pop})^\top X^{pop} (X^\top X)^{-1} X^\top y.$$

Define $\hat{w}$ as the equivalent unit weights such that

$$\hat{\theta}^{PS} = \frac{1}{n} \sum_{i=1}^n \hat{w}_i y_i, \quad \sum_{i=1}^n \hat{w}_i = n.$$

Solving for $\hat{w}$ gives

$$\hat{w} = \frac{n}{N} (N^{pop})^\top X^{pop} (X^\top X)^{-1} X^\top.$$

The outcome y cancels out, so weights depend only on auxiliary variables. This is an advantage, as demographic and geographic information is typically available with minimal missingness. The formula produces S unique equivalent unit weights, one for each poststratification cell. For linear unit weights, renormalization to sum to n is unnecessary unless negative weights occur. Moreover, multiplying the weights by any constant does not affect the estimation of the population mean. Thus, the weights can be renormalized to sum to the known population size, ensuring that $\sum_{i=1}^n \hat{w}_i = \hat{N}$. After this renormalization, the population estimate is

$$\hat{\theta}^{PS} = \frac{\sum_{i=1}^n \hat{w}_i y_i}{\hat{N}},$$

, which corresponds to a form of the Hájek estimator.

Table 2: Summary of linear equivalent weights (simulation example)

	Min	1st Quantile	Median	Mean	3rd Quantile	Max
Value	0.0022	0.15	0.82	1.00	1.52	7.00

2.2 Logistic Regression Equivalent Weights

Let $y = (y_1, \dots, y_n)$ be a binary response and let X be the $n \times k$ matrix of weighting variables. We consider the logistic regression model

$$\Pr(y_i = 1 \mid X_i) = p_i, \quad \text{logit}(p_i) = X_i^\top \beta,$$

with fitted probabilities

$$\hat{p}_i = \sigma(X_i^\top \hat{\beta}), \quad \sigma(t) = \frac{1}{1 + e^{-t}}.$$

For the population poststratification cells X_s^{pop} , $s = 1, \dots, S$, define the model-based population estimator

$$\hat{\theta}^{LR} = \frac{1}{N} \sum_{s=1}^S N_s \sigma(X_s^{\text{pop}\top} \hat{\beta}), \quad N = \sum_{s=1}^S N_s.$$

In survey practice, base weights are generally constructed from prespecified auxiliary variables, such as demographic and geographic characteristics, rather than from the survey outcome. We therefore formulate the equivalent weights in terms of the auxiliary variables and the fitted logistic model. This is also the weighting mechanism used in the data-generating process of our simulation study.

For a linear regression model, the corresponding population estimator can be represented exactly as a weighted average of the observed outcomes. For the logistic model, the mapping from y to $\hat{\theta}^{LR}$ is nonlinear. We therefore consider a local equivalent-weight representation obtained by differentiating the model-based population estimator with respect to the observed responses.

Define the sample and population logistic variance matrices by

$$W(\beta) = \text{diag} \{ \sigma(X\beta) \odot [1 - \sigma(X\beta)] \}$$

and

$$W^{\text{pop}}(\beta) = \text{diag} \{ \sigma(X^{\text{pop}}\beta) \odot [1 - \sigma(X^{\text{pop}}\beta)] \}.$$

In accordance with standard survey-weighting practice, the equivalent weights are treated as functions of the auxiliary variables and the fitted logistic model, rather than as functions of the individual outcomes directly. Thus, conditional on the fitted model,

$$\frac{\partial \hat{w}_i}{\partial y_i} = 0.$$

Under this weighting assumption, the model-based population estimator is represented by the equivalent weighted sample mean,

$$\frac{1}{n} \sum_{i=1}^n \hat{w}_i y_i = \frac{1}{N} \sum_{s=1}^S N_s \sigma(X_s^{\text{pop}\top} \hat{\beta}). \quad (1)$$

The logistic regression estimator satisfies the score equation

$$X^\top (y - \hat{p}) = 0, \quad \hat{p} = \sigma(X\hat{\beta}).$$

Implicit differentiation of the score equation with respect to the observed response vector y gives

$$\frac{d\hat{\beta}}{dy} = (X^\top W(\hat{\beta})X)^{-1} X^\top. \quad (2)$$

Applying the chain rule to the model-based population estimator yields

$$\frac{\partial \hat{\theta}^{LR}}{\partial y^\top} = \frac{1}{N} (N^{\text{pop}})^\top W^{\text{pop}}(\hat{\beta}) X^{\text{pop}} (X^\top W(\hat{\beta})X)^{-1} X^\top, \quad (3)$$

where

$$N^{\text{pop}} = (N_1, \dots, N_S)^\top.$$

Because the weights are treated as fixed with respect to the observed outcomes under the stated

survey-weighting assumption, differentiating the left-hand side of (1) with respect to y_i gives

$$\frac{\partial}{\partial y_i} \left(\frac{1}{n} \sum_{j=1}^n \hat{w}_j y_j \right) = \frac{\hat{w}_i}{n}.$$

Consequently, the local equivalent unit weight is

$$\hat{w}_i = n \frac{\partial \hat{\theta}^{LR}}{\partial y_i}, \quad i = 1, \dots, n. \quad (4)$$

Equivalently, in vector form,

$$\hat{w} = n \left(\frac{\partial \hat{\theta}^{LR}}{\partial y^\top} \right)^\top. \quad (5)$$

Combining (3) and (5) gives the closed-form expression

$$\boxed{\hat{w}(\hat{\beta}) = \frac{n}{N} X(X^\top W(\hat{\beta})X)^{-1} (X^{\text{pop}})^\top W^{\text{pop}}(\hat{\beta}) N^{\text{pop}}.} \quad (6)$$

The definition in (4) gives the local interpretation of the weights. For a sufficiently small perturbation Δy of the observed response vector, a first-order Taylor expansion gives

$$\hat{\theta}^{LR}(y + \Delta y) = \hat{\theta}^{LR}(y) + \frac{1}{n} \hat{w}^\top \Delta y + o(\|\Delta y\|). \quad (7)$$

Thus, $\hat{w}_i/n$ measures the local sensitivity of the model-based population estimator to a perturbation of observation i 's outcome. The vector $\hat{w}$ therefore provides a local equivalent-weight representation of the nonlinear logistic population estimator rather than an exact global representation as a weighted sample mean.

Because the population is represented by S poststratification cells, the resulting weights are determined by the weighting variables and the fitted logistic model. In particular, sampled units with identical weighting-variable patterns receive the same equivalent weight, so there are at most S distinct weights.

In the remainder of the paper, we study these local equivalent weights as a survey-weighting construction, with particular emphasis on their effective sample size and frequentist sampling properties.

For estimation of a population mean, the weights are invariant to a common multiplicative rescaling. We therefore normalize the weights to sum to n when reporting equivalent unit weights. This normalization does not change the resulting weighted mean or effective sample size.

Table 3: Summary of logistic equivalent weights (simulation example)

	Min	1st Quantile	Median	Mean	3rd Quantile	Max
Value	0.0024	0.12	0.83	1.00	1.53	7.01

2.3 Negative and Extreme Weights in Model-based Methods

Model-based equivalent weights can be negative or highly variable, particularly when the sample provides weak information for some poststratification cells or when the model-based population target differs substantially from the sample composition. Negative weights are a known feature of regression-type weighting and should not be interpreted as probabilities.

In the simulations designed to mimic the FFCWS setting, negative weights occur for both linear and logistic equivalent-weight constructions. For the practical comparisons reported below, negative weights are replaced by the minimum positive weight and the resulting weights are renormalized. Extreme positive weights are also trimmed at the 95th percentile in analyses where trimming is explicitly indicated.

These adjustments are post-processing operations. The asymptotic results in Section 3 concern the raw equivalent-weight construction in (6); they do not establish asymptotic properties for the trimmed or negative-weight-adjusted estimator. We therefore report these procedures separately and treat their effects as an empirical sensitivity issue rather than incorporating them into the theoretical estimator.

Ideally, any modification of negative weights would preserve relevant cell-level calibration constraints. Such a procedure is not pursued here because the cells producing negative weights need not overlap with cells containing positive weights. The practical adjustment is therefore intended as a simple stabilization device rather than as a theoretically optimal correction.

Negative weights are not uncommon when constructing weights using model-based approaches. They arise when real-world surveys fail to accurately represent the target population due to over-sampling or undersampling of particular demographic groups, or because of limited sample sizes.

This is expected, as one of the motivations for constructing weights is precisely to correct for imbalances in representativeness.

In most cases, linear equivalent unit weights are positive when samples are randomly drawn from the population. However, in our simulation study designed to mimic the FFCWS data, we encountered negative weights for both linear and logistic equivalent unit weights. To address this, we replace negative weights with the minimum value of the positive weights, an adjustment that has negligible impact on the total weight count across demographic cells once weights are renormalized. Ideally, negative weights would be adjusted within the same cells to preserve cell-specific totals, but in practice this is rarely feasible, as cells with negative weights often do not overlap with cells containing positive weights. Consequently, adjusting for negative weights results in only minor discrepancies in total counts relative to population benchmarks.

2.4 Missingness and Nonresponse

In practice, missing data in demographic and geographic variables used for survey weighting should be assessed before constructing weights. When auxiliary variables are missing, both design-based and model-based weighting become more difficult because the corresponding population and sample cells may no longer be directly comparable.

For nonresponse adjustment, design-based methods typically require an explicit adjustment step based on response rates within adjustment cells. The model-based construction studied here instead uses auxiliary variables directly in the regression model and population poststratification. This does not eliminate nonresponse assumptions: the validity of the model-based estimator still depends on whether the selected weighting variables adequately capture systematic differences relevant to the target population.

In the FFCWS application, city-level population information is not directly available in the CDC natality data and must be constructed using sample information and external population totals. We therefore treat the population cell counts used in the application as fixed inputs to the weighting procedures.

3 Asymptotic Properties

3.1 Finite-population and superpopulation frameworks

The weighting construction is motivated by a finite-population survey problem in which the population cell sizes N_s are treated as known or estimated external benchmarks. The asymptotic results in this section are instead established under a finite-cell superpopulation framework, which provides a tractable large-sample approximation for the estimated weights.

Specifically, we consider a finite set of poststratification cells and assume that the sampled covariates X_i are generated from a distribution that assigns positive probability to the cells relevant to the target population. The asymptotic analysis therefore studies repeated samples from this superpopulation while treating the population-cell structure as fixed. Accordingly, Theorems 1 and 2 should be interpreted as superpopulation results under the stated regularity conditions, rather than as a general design-based asymptotic theory for complex surveys.

The distinction is important. The proposed weights are motivated by survey calibration and poststratification, whereas Theorems 1–2 provide model-based large-sample results under the stated regularity conditions. Extending these results to general complex sampling designs, including clustering and unequal-probability multistage sampling, is an important direction for future work.

Lemma 1 (Uniform constant envelope for logistic equivalent weights with categorical X). *Let X consist entirely of categorical variables encoded as dummies, so that each X_i takes values in a finite set $\mathcal{X}$ and $\sup_{x \in \mathcal{X}} \|x\| < \infty$. Define the closed-form logistic equivalent weights (as a function of β)*

$$\hat{w}(\beta) = \left(\frac{n}{N} (N^{\text{pop}})^\top W^{\text{pop}}(\beta) X^{\text{pop}} ((X^\top W(\beta) X)^{-1} X^\top)^\top \right)^\top,$$

where $W(\beta) = \text{diag}(\sigma(X\beta) \odot (1 - \sigma(X\beta)))$ and $X^{\text{pop}}, N^{\text{pop}}$ are fixed finite matrices/vectors. Assume there exists a neighborhood $\mathcal{N}$ of β_0 and a constant $c > 0$ such that the weighted information matrix $H(\beta) := X^\top W(\beta) X$ satisfies $\lambda_{\min}(H(\beta)) \geq c$ for all $\beta \in \mathcal{N}$. Then there exists $C < \infty$ such that, for all $\beta \in \mathcal{N}$ and all i ,

$$|\hat{w}_i(\beta)| \leq C.$$

Lemma 2 (Continuity of logistic equivalent weights in β). *Under the same assumptions as Lemma 1,*

the map $\beta \mapsto \hat{w}(\beta)$ is continuous on $\mathcal{N}$ (coordinatewise and in any matrix norm).

Theorem 1 (Plug-in LLN for logistic equivalent weights with categorical X). *Let $\{X_i\}_{i=1}^n$ be i.i.d. with law F . Let $\hat{w}_i(\beta)$ denote the closed-form logistic equivalent weights defined in Lemma 1. Assume:*

1. (Envelope) *The assumptions of Lemma 1 hold, so there exists a constant $C < \infty$ such that $|\hat{w}_i(\beta)| \leq C$ for all $\beta \in \mathcal{N}$ and all i (hence $g(x) \equiv C$ is an integrable envelope with $\mathbb{E}g < \infty$ and $\mathbb{E}g^2 < \infty$);*
2. (Continuity) *The assumptions of Lemma 2 hold, so $\beta \mapsto \hat{w}(\beta)$ is continuous on the neighborhood $\mathcal{N}$ of β_0 ;*
3. (Compactness) *$\mathcal{N}$ is compact and $\beta_0 \in \text{int}(\mathcal{N})$.*
4. *$\hat{\beta} \xrightarrow{p} \beta_0$;*

Define

$$\bar{w}_n(\beta) = \frac{1}{n} \sum_{i=1}^n \hat{w}_i(\beta), \quad \overline{w^2}_n(\beta) = \frac{1}{n} \sum_{i=1}^n \hat{w}_i(\beta)^2.$$

Then

$$\bar{w}_n(\hat{\beta}) \xrightarrow{p} \mathbb{E}[\hat{w}_1(\beta_0)], \quad \overline{w^2}_n(\hat{\beta}) \xrightarrow{p} \mathbb{E}[\hat{w}_1(\beta_0)^2].$$

Corollary 1 (ESS Limit). *Let $\text{ESS}_n = (\sum_{i=1}^n w(X_i; \hat{\beta}))^2 / \sum_{i=1}^n w(X_i; \hat{\beta})^2$. Under the assumptions of Theorem 1,*

$$\frac{\text{ESS}_n}{n} = \frac{\bar{w}_n(\hat{\beta})^2}{\overline{w^2}_n(\hat{\beta})} \xrightarrow{p} \frac{(\mathbb{E}[w(X; \beta_0)])^2}{\mathbb{E}[w(X; \beta_0)^2]}.$$

Proof. Apply Theorem 1 to $\bar{w}_n(\hat{\beta})$ and $\overline{w^2}_n(\hat{\beta})$, then invoke the continuous mapping theorem. $\square$

Remark 1 (Scale invariance). If $\tilde{w} = cw$ for any constant $c > 0$ (e.g., normalization), then ESS_n/n is unchanged since both numerator and denominator are multiplied by c^2 .

Lemma 3 (Differentiability and uniform derivative envelope). *Under the assumptions of Lemmas 1–2, the map $\beta \mapsto \hat{w}(\beta)$ is continuously differentiable (C^1) on $\mathcal{N}$. Moreover, there exists a finite constant C' (depending only on $X, X^{\text{pop}}, N^{\text{pop}}, c$) such that*

$$\sup_{\beta \in \mathcal{N}} \|\nabla_{\beta} \hat{w}_i(\beta)\| \leq C' \quad \text{for all } i.$$

Theorem 2 (Asymptotic linearity of $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \hat{w}_i(\hat{\beta}) Y_i$). Let $\hat{w}_i(\beta)$ be the logistic equivalent weights. Assume:

- (i) (Constant envelope for weights) Lemma 1 holds: $|\hat{w}_i(\beta)| \leq C$ for all $\beta \in \mathcal{N}$ and all i .
- (ii) (C^1 and uniform derivative envelope) Lemma 3 holds: $\beta \mapsto \hat{w}(\beta)$ is C^1 on $\mathcal{N}$ and there exists $C' < \infty$ such that $\sup_{\beta \in \mathcal{N}} \|\nabla_{\beta} \hat{w}_i(\beta)\| \leq C'$ for all i .
- (iii) (Logistic MLE regularity) The logistic model is correctly specified at β_0 with $p_0(x) = \sigma(x^{\top} \beta_0)$; β_0 is an interior point of the parameter space; X has $\mathbb{E}\|X\|^2 < \infty$; and the Fisher information $\mathcal{I}(\beta_0) = \mathbb{E}[X X^{\top} p_0(X) \{1 - p_0(X)\}]$ is positive definite.

Define $g(\beta) = \mathbb{E}[\hat{w}_1(\beta) Y_1]$ and

$$A := \nabla_{\beta} g(\beta) \Big|_{\beta=\beta_0} = \mathbb{E}[\nabla_{\beta} \hat{w}_1(\beta_0) Y_1] \quad (\text{under correct logit spec., } = \mathbb{E}[\nabla_{\beta} \hat{w}_1(\beta_0) p_0(X_1)]).$$

Then

$$\sqrt{n} (\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \underbrace{\left\{ \hat{w}_i(\beta_0) Y_i - \mathbb{E}[\hat{w}_1(\beta_0) Y_1] \right\}}_{\psi_1(Z_i)} + A^{\top} \sqrt{n} (\hat{\beta} - \beta_0) + o_p(1),$$

so that

$$\sqrt{n} (\hat{\theta} - \theta_0) \Rightarrow \mathcal{N}(0, V), \quad V = \mathbb{E}[\psi(Z)^2],$$

with influence function

$$\psi(Z) = \left\{ \hat{w}(\beta_0) Y - \mathbb{E}[\hat{w}(\beta_0) Y] \right\} + A^{\top} \mathcal{I}^{-1} X \{Y - p_0(X)\}.$$

A consistent plug-in variance is

$$\hat{V} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2, \quad \hat{\psi}_i = \hat{w}_i(\hat{\beta}) Y_i - \bar{m}_n + \hat{A}^{\top} \hat{\mathcal{I}}^{-1} X_i \{Y_i - \hat{p}_i\},$$

where

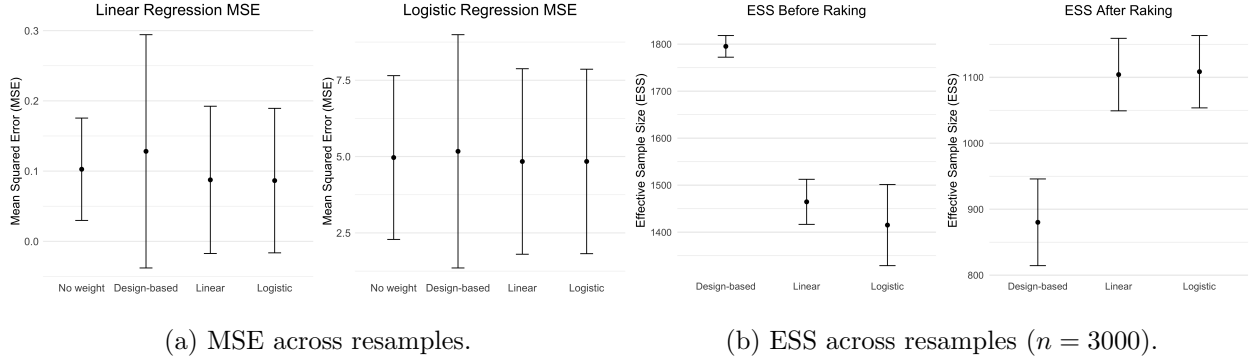
$$\bar{m}_n = \frac{1}{n} \sum_{j=1}^n \hat{w}_j(\hat{\beta}) Y_j, \quad \hat{p}_i = \sigma(X_i^{\top} \hat{\beta}), \quad \hat{\mathcal{I}} = \frac{1}{n} \sum_{j=1}^n X_j X_j^{\top} \hat{p}_j (1 - \hat{p}_j), \quad \hat{A} = \frac{1}{n} \sum_{j=1}^n \nabla_{\beta} \hat{w}_j(\hat{\beta}) \hat{p}_j.$$

4 Results

4.1 Simulation Results

We first examine the performance of the weighting methods in regression modeling and their effect on effective sample size. Figure 1a reports the pooled mean squared error (MSE) of the regression coefficients across 300 independent resamples. Both model-based approaches—linear and logistic weighting—achieve lower MSE than the design-based and unweighted regressions, indicating greater stability in the regression estimates under the simulated setting.

Figure 1b presents the effective sample size (ESS) of the weights across the 300 replications before and after raking. The design-based method has the highest ESS before raking, followed by the linear and logistic weights. After raking, the model-based approaches, particularly logistic weighting, preserve ESS more effectively, whereas the design-based weights experience a more pronounced reduction. These results provide a finite-sample assessment of the behavior of the estimated weights beyond their performance in estimating the population mean.



Next, we evaluate the finite-sample performance of the three equivalent-weight estimators in estimating the population mean of a binary outcome. The simulation target is the superpopulation mean

$$\theta_0 = \frac{1}{N} \sum_{j=1}^N p_j,$$

where p_j denotes the true outcome probability used to generate the binary responses. Table 4 reports the Monte Carlo mean of the estimator, bias, mean squared error (MSE), and empirical standard deviation.

Table 4: Finite-sample performance of equivalent-weight estimators

Method	Estimate	Bias	MSE	Empirical SD
Design-based	0.352	-0.021	0.000595	0.0121
Linear	0.374	0.001	0.000148	0.0121
Logistic	0.369	-0.004	0.000194	0.0134
Population mean			0.373	

The linear estimator has the smallest MSE in this simulation and is approximately unbiased, while the logistic estimator also exhibits small finite-sample bias and competitive MSE. The design-based estimator has the largest bias and MSE under the data-generating process considered. These results illustrate that the proposed logistic equivalent weights provide competitive finite-sample point estimation, although they do not uniformly dominate the alternative weighting methods.

To further connect the finite-sample evidence with the asymptotic results, we next examine the effective sample size implied by the estimated weights. The effective sample size is defined as

$$\text{ESS} = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2}.$$

To evaluate Theorem 1 and Corollary 1, we compare the empirical ESS ratio $\bar{w}_n(\hat{\beta})^2/\overline{w_n^2}(\hat{\beta})$ with the population target $\mathbb{E}[w]^2/\mathbb{E}[w^2]$ using 300 replications. In practice, we compute the effective sample size ratio using sample proportions,

$$\frac{\left(\sum_{s=1}^S \hat{\pi}_s w(x_s; \beta_0)\right)^2}{\sum_{s=1}^S \hat{\pi}_s w(x_s; \beta_0)^2}, \quad \hat{\pi}_s = \frac{n_s}{n},$$

as a finite-sample analogue of the population quantity

$$\frac{\mathbb{E}[w(X; \beta_0)]^2}{\mathbb{E}[w(X; \beta_0)^2]} = \frac{\left(\sum_{s=1}^S \pi_s w(x_s; \beta_0)\right)^2}{\sum_{s=1}^S \pi_s w(x_s; \beta_0)^2}, \quad \pi_s = \frac{N_s}{N}.$$

Since $\hat{\pi}_s \rightarrow \pi_s$ as $n \rightarrow \infty$, the sample-based ratio converges to the population limit in Corollary 1. Table 5 reports the mean and interquartile range of two finite-sample analogues of the ESS ratio across 300 replications. The first row, “empirical ESS ratio,” is computed directly from the individual fitted weights as $\bar{w}_n(\hat{\beta})^2/\overline{w_n^2}(\hat{\beta})$. The second row, “cell-based ESS ratio,” evaluates the population-style expression using sample proportions $\hat{\pi}$ in place of π . The two ratios are close, with

means of 0.38 and 0.37, respectively, providing finite-sample evidence consistent with the population ESS limit in Corollary 1. For $n = 3000$, the empirical ESS ratio of 0.38 corresponds to an effective sample size of approximately 1,140.

Table 5: Finite-sample ESS/ n ratios: mean and IQR

	Mean	IQR [Q1, Q3]
Empirical ESS ratio	0.38	[0.37, 0.39]
Cell-based ESS ratio	0.37	[0.36, 0.38]

Finally, we evaluate the finite-sample behavior of the asymptotic linear representation in Theorem 2. We compute the centered and scaled estimator

$$\sqrt{n}(\hat{\theta} - \theta_0)$$

across 300 independent replications. The distribution of the centered and scaled estimator is approximately Gaussian, with only mild deviations in the tails, as shown by the normal Q–Q plot in Figure 2.

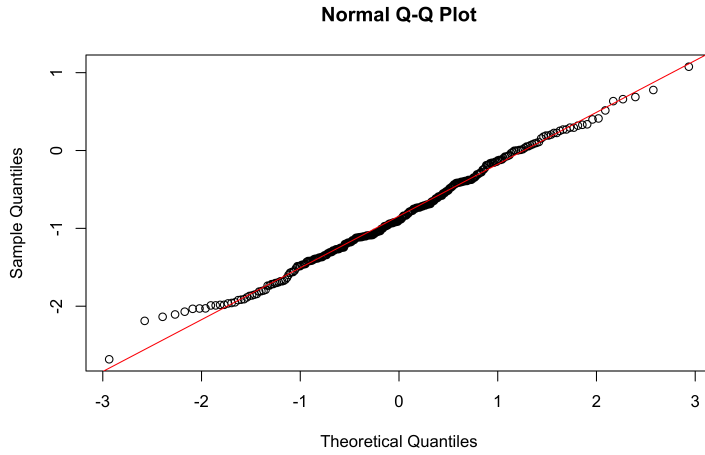


Figure 2: Normal Q–Q plot of the centered and scaled estimator $\sqrt{n}(\hat{\theta} - \theta_0)$ across 300 replications.

We next evaluate the plug-in variance estimator from Theorem 2. For asymptotic variance estimation, we use the unnormalized raw equivalent weights when computing the gradient term $\hat{A}$, while the equivalent weights are normalized to sum to n in the weighted-mean component of

the influence function. This normalization ensures that $\hat{\theta}$ is expressed on the $1/n$ scale required in Theorem 2. Across the 300 replications, the estimated asymptotic variance has mean approximately 0.55 and interquartile range approximately $[0.53, 0.57]$. The empirical Monte Carlo asymptotic variance is approximately 0.54, indicating close agreement between the plug-in variance estimator and the observed sampling variability.

For comparison, we also construct Wald confidence intervals for the linear equivalent-weight estimator using its standard influence-function variance estimator. Table 6 reports the empirical Monte Carlo variance, mean estimated variance, mean standard error, and empirical coverage of nominal 95% confidence intervals.

Table 6: Variance estimation and confidence-interval performance

Method	Empirical V	Mean $\hat{V}$	Mean SE	95% Coverage
Linear	0.437	0.479	0.0126	97.3%
Logistic	0.543	0.554	0.0136	94.7%

For the logistic estimator, the mean plug-in variance estimate (0.554) is close to the empirical Monte Carlo variance (0.543). The resulting nominal 95% Wald confidence intervals achieve 94.7% empirical coverage, indicating approximately nominal finite-sample inference in this simulation. The linear estimator also shows good coverage at 97.3%, although its plug-in variance estimate is somewhat more conservative in this setting.

4.2 Applications to FFCWS Study

We apply the weighting methods to the FFCWS using $n = 3,442$ sampled units representing a population of 1,131,308 individuals. Both model-based methods produce negative weights in this application; for the practical comparisons reported here, we replace negative weights with the smallest positive weight and renormalize. The logistic model uses father interview status as the binary response for constructing the weights because this variable has no missing values. This response is used only to construct the weights; the substantive outcome analyzed below is CPS contact. When raking the model-based weights, we use the same four demographic variables as in the national FFCWS weights, excluding city. We refer to the national FFCWS weights as the design-based weights, which are already raked in the FFCWS data.

The population-count results in Table 7 show that the linear and logistic weighting procedures

provide broadly similar approximations to the population totals, although both differ from the design-based totals in some education categories. Table 8 shows that the logistic weights have the largest ESS among the model-based methods, both before and after raking. After raking, the logistic weights retain a larger fraction of their pre-raking ESS than the design-based weights.

Table 7: Population counts of live births by education

Education	<8 grade	Some HS	HS	Some College	College+	Total
Design-raked	111,324	211,988	340,211	214,319	253,467	1,131,309
Linear model	73,232	267,083	367,476	242,049	181,469	1,131,309
Logistic model	60,214	207,301	316,517	258,784	288,491	1,131,309

Table 8: Effective Sample Size (ESS) for FFCWS Weights ($n = 3442$)

	Design-based	Linear	Logit	Linear-raked	Logit-raked
ESS	527	1273	1435	1082	1301

Using the logistic equivalent weights, the estimated prevalence of CPS contact in the FFCWS is $\hat{\theta} = 0.49$ with $\hat{V} = 0.42$, yielding a 95% confidence interval of $[0.47, 0.51]$ under the plug-in variance calculation. This illustrates how the proposed variance estimator can be used with a single dataset without bootstrap or replication-based variance estimation. Because the weighting response and substantive outcome are different variables in this application, the estimate should be interpreted as an illustration of the weighting procedure rather than as evidence that father interview status is itself a sufficient model for CPS-contact prevalence.

To examine subgroup behavior, we compare CPS-contact prevalence across cities in Figure 3. The weighting methods generally reduce the unweighted estimates and move them toward the external prevalence range used as a benchmark (Berger et al., 2025), but the magnitude of the adjustment varies by city. The design-based and linear weights produce relatively consistent adjustments, whereas the logistic weights show larger changes in several cities. Thus, in this application the choice of weighting method matters most for particular subgroups rather than uniformly across the population.

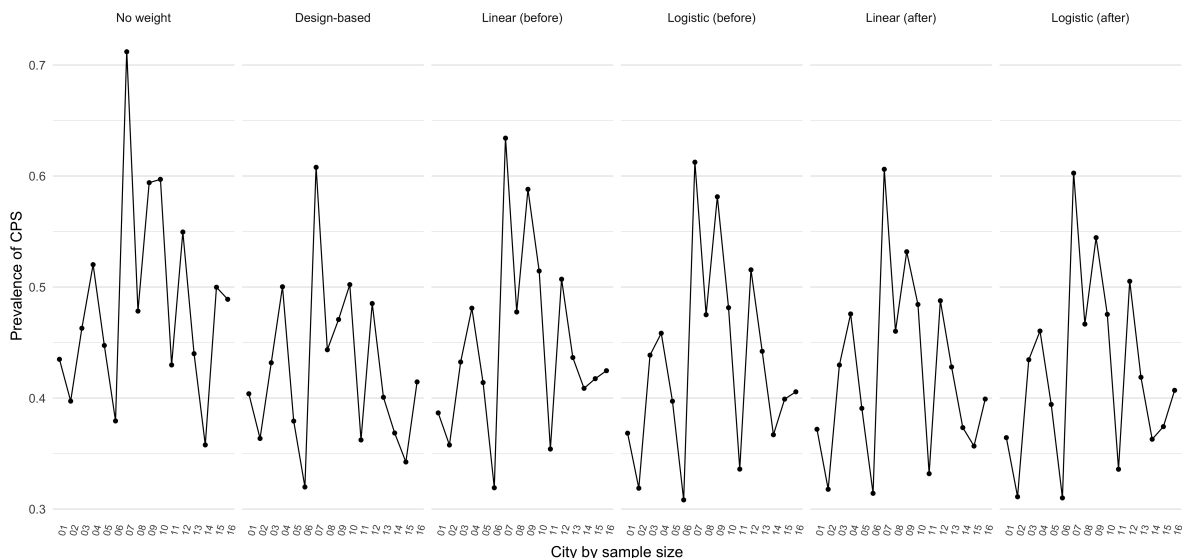


Figure 3: Prevalence of Child Protective Services (CPS) contact by city before and after raking

5 Discussion

5.1 Key Findings

This study develops and evaluates logistic regression equivalent weights as a model-based survey-weighting construction. The central methodological object is a local equivalent-weight representation of the logistic population estimator, with the weights defined as scaled first derivatives of the model-based population estimator with respect to the observed outcomes. Unlike the linear regression case, the logistic estimator is nonlinear in the observed outcomes, so the proposed weights should be interpreted as a local linear approximation to the model-based population estimator rather than as an exact global identity.

The theoretical results establish convergence of the empirical ESS ratio to a population limit under a finite-cell superpopulation framework. We also derive an asymptotic linear representation for the weighted estimator and a plug-in variance estimator. The simulation study provides evidence that these approximations can be useful at the sample sizes considered here: the empirical distribution of the centered and scaled estimator is close to the Gaussian approximation, and the plug-in variance estimates are close to the empirical variance.

The simulation results also show that no single weighting method dominates across all criteria.

Before raking, the design-based weights have the largest ESS in the reported simulation, whereas after raking the model-based weights, particularly the logistic weights, retain a larger fraction of their ESS. In terms of population recovery, the linear model is often closest to the simulated population benchmarks, while the logistic model gives similar but not uniformly better results. These findings suggest that logistic weighting may be attractive when ESS retention is important, but they do not establish universal superiority over design-based or linear weighting.

In the FFCWS application, the logistic equivalent weights produce the highest ESS among the methods considered, both before and after raking. The weighting procedures also change estimated CPS-contact prevalence across cities, with the magnitude of the adjustment varying across subgroups. The linear weights appear more stable across cities, whereas the logistic weights show greater sensitivity in some locations. This illustrates a practical trade-off between stability and flexibility.

5.2 Limitations

Several limitations should be emphasized. First, the proposed logistic weights provide a local representation of a nonlinear model-based estimator rather than an exact global representation by a weighted sample mean. Their interpretation and inferential properties therefore depend on the local approximation underlying the construction.

Second, the theoretical results concern the raw equivalent weights. Trimming extreme weights and replacing negative weights are practical post-processing steps used in some empirical analyses, but they are not covered by the stated asymptotic theory. Their effect on bias, variance, and calibration deserves separate study.

Third, the current simulations do not provide a comprehensive analysis of model misspecification. The logistic model can perform differently when the relationship between the outcome and the weighting variables is misspecified, when interactions are omitted, or when some poststratification cells are sparse. A systematic misspecification study would be an important extension.

Fourth, the current theoretical framework uses a finite-cell superpopulation approximation rather than a full design-based asymptotic treatment of complex multistage sampling. The behavior of the proposed weights under clustering, unequal-probability sampling, and design effects therefore requires additional investigation.

Finally, the choice of weighting variables and the availability of population cell counts remain central to the method. The approach does not automatically incorporate richer auxiliary population information beyond the poststratification structure supplied to the model.

5.3 Relationship to Recent Work

Recent work by Giordano et al. (2026) develops locally equivalent weights for multilevel regression and poststratification (MrP), including nonlinear outcome models such as logistic regression. Their framework derives an asymptotic closed-form expression for logistic locally equivalent weights from a Bayesian MrP formulation using Bernstein–von Mises and delta-method approximations. Our derivation proceeds independently from a frequentist perspective, starting from the logistic estimating equations and applying the chain rule to the model-based population estimator. Under categorical poststratification, this leads to the same closed-form weighting expression. Our focus, however, is on the interpretation and use of these weights as survey weights and on their frequentist sampling properties. We therefore study their effective sample size, asymptotic linearity, plug-in variance estimation, and finite-sample behavior.

5.4 Future Research

Several extensions are natural. First, the theoretical results could be generalized to complex survey designs, including stratified multistage sampling, clustering, and unequal-probability sampling. Second, a systematic study of model misspecification could clarify when logistic weighting provides an advantage over linear or design-based approaches. Third, methods for stabilizing extreme and negative weights could be developed with explicit calibration and inferential guarantees.

A further direction is to incorporate richer population information through hierarchical and nonparametric models. In particular, Gaussian-process and hierarchical Bayesian approaches can provide flexible representations of population-level relationships while allowing uncertainty about sparse cells to be propagated through the weighting procedure. Such extensions motivate the broader research direction developed in our subsequent work.

A Supplementary One-Unit Perturbation Analysis

This appendix provides a supplementary numerical assessment of the sensitivity of the logistic equivalent weights to individual outcome perturbations. This analysis is not required for the derivation of the equivalent weights in Section 2 or for the asymptotic results. Instead, it examines the extent to which the fitted weights may change when a single observed binary outcome is perturbed.

A.1 One-Unit Perturbation Analysis

Let $y^{(-i)}$ denote the response vector obtained by changing an observed value $y_i = 1$ to $y_i = 0$, while leaving all other outcomes unchanged. Equivalently,

$$y^{(-i)} = y - e_i,$$

where $e_i = (0, \dots, 0, 1, 0, \dots, 0)^\top$ has its 1 in position i .

For each perturbation, the logistic regression model is re-estimated using $y^{(-i)}$, and the equivalent weights are recomputed. Denote the resulting weight vector by

$$\hat{w}^{(-i)} = \hat{w}(y^{(-i)}),$$

while $\hat{w} = \hat{w}(y)$ denotes the weights obtained from the original observed response vector.

A first-order finite-difference approximation to the derivative of the weight vector with respect to y_i is

$$\frac{\partial \hat{w}(y)}{\partial y_i} \approx \frac{\hat{w}(y^{(-i)}) - \hat{w}(y)}{y_i^{(-i)} - y_i} = \hat{w}(y) - \hat{w}(y^{(-i)}), \quad (8)$$

because $y_i^{(-i)} - y_i = -1$. Thus, the change in the fitted weight vector resulting from changing one observed outcome from 1 to 0 provides a direct numerical approximation to its local outcome sensitivity.

Because y is binary, the Jacobian–vector product can be written as

$$(\nabla \hat{w})y = \sum_{i: y_i=1} \frac{\partial \hat{w}(y)}{\partial y_i}.$$

Using the one-unit perturbations in (8) gives the finite-difference approximation

$$(\nabla \hat{w})y \approx \sum_{i: y_i=1} \left\{ \hat{w}(y) - \hat{w}(y^{(-i)}) \right\}. \quad (9)$$

This calculation provides a direct numerical assessment of the magnitude of the outcome-dependent component of the fitted weights.

A.2 Empirical Sensitivity Assessment

Across the simulation experiments, the outcome-dependent correction term $(\nabla \hat{w})y$ was extremely small. The observed values ranged from approximately -1.4×10^{-4} to 4.2×10^{-5} , with mean approximately -4.0×10^{-6} and median approximately -2.4×10^{-6} .

In a separate one-unit perturbation check, changing a single binary outcome altered

$$(\nabla \hat{w})y + \hat{w}$$

by at most approximately 0.02, with a median change of approximately 0.002.

These calculations provide a numerical sensitivity check for the settings considered. They suggest that the indirect dependence of the fitted equivalent weights on individual outcomes through the fitted logistic model is small in the simulation and empirical settings examined. They are not intended to establish that outcome dependence is asymptotically negligible under all sampling designs or model configurations, and they are not required for the derivation of the equivalent weights.

B Proofs

B.1 Proof of Lemma 1

Proof. The logistic variance satisfies $\sigma(t)\{1 - \sigma(t)\} \leq 1/4$ for all t , so $\|W(\beta)\|_{\text{op}} \leq 1/4$. The matrices X^{pop} and N^{pop} are fixed, hence have finite operator norms. Because $H(\beta)$ is symmetric

positive definite with $\lambda_{\min}(H(\beta)) \geq c$, its spectral norm satisfies

$$\|H(\beta)^{-1}\| = \frac{1}{\lambda_{\min}(H(\beta))} \leq \frac{1}{c}, \quad (10)$$

for all $\beta \in \mathcal{N}$. Since $X_i \in \mathcal{X}$ takes values in a finite set of dummy vectors, $\|X\|$ and $\|X^\top\|$ are uniformly bounded. Thus in the general form one might write

$$|\hat{w}_i(\beta)| \leq C(1 + \|X_i\|^{m'}),$$

but because $\|X_i\|$ is uniformly bounded this polynomial term can be absorbed into a constant, yielding the sharper bound

$$|\hat{w}_i(\beta)| \leq C',$$

for some finite C' depending only on $(X, X^{\text{pop}}, N^{\text{pop}}, c)$. $\square$

B.2 Proof of Lemma 2

Proof. The logistic map $\sigma(t) = 1/(1 + e^{-t})$ is infinitely differentiable on $\mathbb{R}$, hence continuous. Therefore $\beta \mapsto \sigma(X^{\text{pop}}\beta)$ and $\beta \mapsto W(\beta)$ are continuous. Matrix multiplication and the Hadamard product are continuous operations, so $\beta \mapsto X^\top W(\beta)X = H(\beta)$ is continuous. Now, because $\lambda_{\min}(H(\beta)) \geq c > 0$ for all $\beta \in \mathcal{N}$, the set $\{H(\beta) : \beta \in \mathcal{N}\}$ consists entirely of nonsingular matrices bounded away from singularity. On this set, the matrix inversion map $A \mapsto A^{-1}$ is continuous. Therefore $\beta \mapsto H(\beta)^{-1}$ is continuous on $\mathcal{N}$. Finally, composing continuous maps and multiplying on the left and right by the fixed matrices X^{pop} and $(N^{\text{pop}})^\top$ preserves continuity. Hence $\beta \mapsto \hat{w}(\beta)$ is continuous on $\mathcal{N}$. $\square$

B.3 Proof of Theorem 1

Proof. Write

$$\bar{w}_n(\hat{\beta}) - \mathbb{E}[\hat{w}_1(\beta_0)] = \underbrace{\left(\bar{w}_n(\hat{\beta}) - \mathbb{E}[\hat{w}_1(\hat{\beta})]\right)}_{(A)} + \underbrace{\left(\mathbb{E}[\hat{w}_1(\hat{\beta})] - \mathbb{E}[\hat{w}_1(\beta_0)]\right)}_{(B)}.$$

(A) Uniform LLN. Consider the class $\mathcal{F} = \{\hat{w}(\cdot; \beta) : \beta \in \mathcal{N}\}$. By Lemma 2, $\beta \mapsto \hat{w}(\beta)$ is

continuous; by Lemma 1, $|\hat{w}_i(\beta)| \leq C$ for all $\beta \in \mathcal{N}$, so $g \equiv C$ is an integrable envelope. With continuity and compact $\mathcal{N}$, $\mathcal{F}$ is totally bounded in $L_1(F)$ and, with envelope g , is Glivenko–Cantelli. By the uniform weak law of large numbers (Wald, 1949; van der Vaart, 1998),

$$\sup_{\beta \in \mathcal{N}} \left| \bar{w}_n(\beta) - \mathbb{E}[\hat{w}_1(\beta)] \right| \xrightarrow{P} 0.$$

Since $\hat{\beta} \xrightarrow{P} \beta_0 \in \text{int}(\mathcal{N})$, we have (A) $\xrightarrow{P} 0$.

(B) Dominated convergence. By Lemma 2, $\hat{w}_1(\hat{\beta}) \rightarrow \hat{w}_1(\beta_0)$ almost surely as $\hat{\beta} \rightarrow \beta_0$, and by Lemma 1, $|\hat{w}_1(\hat{\beta})| \leq g(X_1) \equiv C$ with $\mathbb{E}[g] < \infty$. Therefore, by dominated convergence, $\mathbb{E}[\hat{w}_1(\hat{\beta})] \rightarrow \mathbb{E}[\hat{w}_1(\beta_0)]$, so (B) $\rightarrow 0$.

Combining (A) and (B) gives $\bar{w}_n(\hat{\beta}) \xrightarrow{P} \mathbb{E}[\hat{w}_1(\beta_0)]$.

The argument for $\overline{w}_n^2(\hat{\beta})$ is identical, applied to the class $\{\hat{w}(\cdot; \beta)^2 : \beta \in \mathcal{N}\}$ with envelope $g^2 \equiv C^2$ (and $\mathbb{E}[g^2] < \infty$), yielding $\overline{w}_n^2(\hat{\beta}) \xrightarrow{P} \mathbb{E}[\hat{w}_1(\beta_0)^2]$. $\square$

B.4 Proof of Lemma 3

Proof. By Lemmas 1 and 2, $\hat{w}(\beta)$ is bounded and continuous on $\mathcal{N}$. Since the logistic map σ is C^∞ , we have $W(\beta) = \text{diag}(\sigma(X\beta) \odot (1 - \sigma(X\beta))) \in C^1$ and $H(\beta) = X^\top W(\beta)X \in C^1$. On $\mathcal{N}$, $\lambda_{\min}(H(\beta)) \geq c > 0$, hence the inversion map is C^1 with

$$D(A^{-1})[\Delta] = -A^{-1}\Delta A^{-1} \quad (\text{Fréchet derivative}). \quad (11)$$

Step 1: Bound $D(H^{-1})$. For a direction $h \in \mathbb{R}^p$,

$$DH(\beta)[h] = X^\top DW(\beta)[h]X,$$

and, writing $p_i = \sigma(x_i^\top \beta)$,

$$(DW(\beta)[h])_{ii} = p_i(1 - p_i)(1 - 2p_i)(x_i^\top h).$$

Hence $\|DW(\beta)[h]\| \leq \frac{1}{4} \|X\| \|h\|$, and by submultiplicativity

$$\|DH(\beta)[h]\| \leq C_X \|h\|, \quad C_X := \frac{1}{4} \|X^\top\| \|X\|^2. \quad (12)$$

Using (11),

$$D(H(\beta)^{-1})[h] = -H(\beta)^{-1} DH(\beta)[h] H(\beta)^{-1},$$

so

$$\|D(H(\beta)^{-1})[h]\| \leq \|H(\beta)^{-1}\|^2 \|DH(\beta)[h]\| \leq \|H(\beta)^{-1}\|^2 C_X \|h\|, \quad (13)$$

with

$$\|H(\beta)^{-1}\| = \frac{1}{\lambda_{\min}(H(\beta))} \leq \frac{1}{c}, \quad (14)$$

and combining (13)–(14) yields

$$\|D(H(\beta)^{-1})[h]\| \leq c^{-2} C_X \|h\|. \quad (15)$$

Step 2: Bound $D\hat{w}(\beta)[h]$. Write $\hat{w}(\beta) = \frac{n}{N} T(\beta)^\top S(\beta)$ with

$$T(\beta) := H(\beta)^{-1} X^\top, \quad S(\beta) := X^{\text{pop}\top} W^{\text{pop}}(\beta) N^{\text{pop}}.$$

By product rule,

$$D\hat{w}(\beta)[h] = \frac{n}{N} \left(DT(\beta)[h]^\top S(\beta) + T(\beta)^\top DS(\beta)[h] \right).$$

Using (15),

$$\|DT(\beta)[h]\| \leq c^{-2} C_X \|X^\top\| \|h\|, \quad \|T(\beta)\| \leq \|H(\beta)^{-1}\| \|X^\top\| \leq c^{-1} \|X^\top\|.$$

Since $X^{\text{pop}}, N^{\text{pop}}$ are fixed and $\sigma(u)(1 - \sigma(u)) \leq \frac{1}{4}$, there exist finite constants C_1, C_2 (depending only on $X^{\text{pop}}, N^{\text{pop}}$ and logistic bounds) such that

$$\|S(\beta)\| \leq C_1, \quad \|DS(\beta)[h]\| \leq C_2 \|h\|.$$

Therefore, by submultiplicativity,

$$\|D\hat{w}(\beta)[h]\| \leq \frac{n}{N} \left((c^{-2}C_X\|X^\top\|)C_1 + (c^{-1}\|X^\top\|)C_2 \right) \|h\| =: C' \|h\|. \quad (16)$$

Step 3: From directional to gradient bound (each coordinate). For the i -th coordinate $f(\beta) = \hat{w}_i(\beta)$ (a scalar),

$$|D\hat{w}_i(\beta)[h]| \leq \|D\hat{w}(\beta)[h]\| \leq C' \|h\|.$$

Taking $\sup_{\|h\|=1}$ gives

$$\|\nabla_\beta \hat{w}_i(\beta)\| = \sup_{\|h\|=1} |D\hat{w}_i(\beta)[h]| \leq C' \quad \text{for all } \beta \in \mathcal{N} \text{ and all } i,$$

which is the desired uniform derivative envelope. Finally, since all components used are C^1 and $H(\beta)^{-1}$ is C^1 on $\mathcal{N}$, it follows that $\beta \mapsto \hat{w}(\beta)$ is C^1 as well (part (i)). This completes the proof of the lemma. $\square$

B.5 Proof of Theorem 2

Proof. Write

$$\hat{\theta} = P_n\{\hat{w}(\hat{\beta})Y\}, \quad \theta_0 = P\{\hat{w}(\beta_0)Y\},$$

hence

$$\hat{\theta} - \theta_0 = \underbrace{\{P_n - P\}[\hat{w}(\beta_0)Y]}_{(I)} + \underbrace{\left(P\hat{w}(\hat{\beta})Y - P\hat{w}(\beta_0)Y\right)}_{(II)} + \underbrace{\left\{(P_n\hat{w}(\hat{\beta})Y - P\hat{w}(\hat{\beta})Y) - (P_n\hat{w}(\beta_0)Y - P\hat{w}(\beta_0)Y)\right\}}_{(III)}.$$

(I) By Lemma 1, $\hat{w}(\beta_0)Y$ has an integrable constant envelope; the i.i.d. CLT gives

$$\sqrt{n}(I) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\hat{w}_i(\beta_0)Y_i - P[\hat{w}(\beta_0)Y]\} \Rightarrow \mathcal{N}(0, \mathbb{V}[\hat{w}(\beta_0)Y]).$$

(II) By Lemma 3, $\beta \mapsto \hat{w}(\beta)$ is C^1 on $\mathcal{N}$; therefore $g(\beta) = P[\hat{w}(\beta)Y]$ is differentiable at β_0 with gradient $A = P[\nabla_\beta \hat{w}(\beta_0)Y]$. A first-order expansion yields $g(\hat{\beta}) - g(\beta_0) = A^\top(\hat{\beta} - \beta_0) + o_p(\|\hat{\beta} - \beta_0\|)$. By (iii) and the influence function representation of the MLE in logistic regression (Hampel et al.,

1986; van der Vaart, 1998, Thm. 5.23),

$$\begin{aligned}\sqrt{n}(II) &= A^\top \sqrt{n}(\hat{\beta} - \beta_0) + o_p(1) \\ &= A^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathcal{I}^{-1} X_i \{Y_i - p_0(X_i)\} \right) + o_p(1).\end{aligned}$$

(III) By Lemma 1, the class $\mathcal{F} := \{\hat{w}(\cdot; \beta)Y : \beta \in \mathcal{N}\}$ has a constant envelope. By Lemma 3, $\beta \mapsto \hat{w}(\beta)$ is C^1 on $\mathcal{N}$ with $\sup_{\beta \in \mathcal{N}} \|\nabla_\beta \hat{w}_i(\beta)\| \leq C'$, so the parametrization $\beta \mapsto \hat{w}(\cdot; \beta)Y$ is Lipschitz in β , uniformly on $\mathcal{N}$. Restricting to a compact neighborhood of β_0 , the class is finite-dimensional, uniformly bounded, and $L^2(P)$ -Lipschitz in β , and is therefore P -Donsker. Since we also have $\|\hat{w}(\cdot; \hat{\beta}) - \hat{w}(\cdot; \beta_0)\|_{L^2(P)} \xrightarrow{P} 0$, by (van der Vaart, 1998, Lemma 19.24),

$$\sqrt{n}(III) = \sqrt{n}(P_n - P)[\hat{w}(\hat{\beta})Y - \hat{w}(\beta_0)Y] = o_p(1).$$

Combining (I)–(III) gives the stated linear expansion and asymptotic variance. Consistency of $\hat{V}$ follows by LLN and continuous mapping. $\square$

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